

A small-ball metric space whose strongly norm-attaining maps contain ℓ_2

Candidate full-counterexample packet for arXiv:2202.06855, Question 4.3

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Status: candidate full counterexample, likely valid, subject to human review.
There is a pointed metric space M with the small ball property for which $\text{SNA}(M)$ contains a closed subspace isometric to ℓ_2 . Hence the polyhedrality assertion in Question 4.3 is false.

1 The question and the answer

Kadets and Roldán prove that if M is σ -precompact, then every Banach subspace of $\text{SNA}(M)$ is separable and isomorphically polyhedral. Question 4.3 asks whether the same conclusion follows from the weaker small ball property [1]. The answer is no. In fact the counterexample works uniformly over all separable infinite-dimensional reflexive Banach spaces.

Theorem 1. *Let X be a separable infinite-dimensional reflexive real Banach space. There is a dense pointed metric subspace $M \subset X$ with the small ball property such that the linear map*

$$J : X^* \longrightarrow \text{Lip}_0(M), \quad (Jx^*)(x) = x^*(x) - x^*(o),$$

where $o \in M$ is fixed, is an isometric embedding and $J(X^) \subset \text{SNA}(M)$.*

In particular, for $X = \ell_2$, the set $\text{SNA}(M)$ contains a closed subspace isometric to ℓ_2 , which is not isomorphically polyhedral.

2 A dense small-ball set has every vector as a chord

We include the short construction underlying Proposition 5.1 of Behrends–Kadets [2]; this makes the counterexample self-contained.

Lemma 1. *Every separable Banach space X has a dense G_δ subset M with the small ball property. Moreover, every such dense G_δ set satisfies $M - M = X$.*

Proof. Choose a dense sequence (x_n) in X and positive numbers $a_n \rightarrow 0$. Put

$$M = \bigcap_{k=1}^{\infty} \bigcup_{n=1}^{\infty} B^o(x_n, a_{k+n}). \tag{1}$$

For each k , the displayed union is open and contains the dense set $\{x_n : n \geq k\}$, so it is dense. Since X is complete, the Baire category theorem makes M a dense G_δ set.

Given $\varepsilon_0 > 0$, choose k so far out that $a_{k+n} < \varepsilon_0$ for every $n \geq 1$. Equation (1) gives

$$M \subset \bigcup_{n=1}^{\infty} B(x_n, a_{k+n}),$$

where all radii are smaller than ε_0 and tend to zero. Thus M has the small ball property.

Finally fix $u \in X$. Both M and $M - u$ are dense G_δ subsets of the complete space X . Their intersection is nonempty. If $q \in M \cap (M - u)$, then $q \in M$ and $q + u \in M$, so $u = (q + u) - q \in M - M$. $\square$

3 Every functional strongly attains on a chord

Proof of Theorem 1. Take the set M from Lemma 1 and choose a base point $o \in M$. For $x^* \in X^*$ the function Jx^* vanishes at o and

$$|(Jx^*)(p) - (Jx^*)(q)| = |x^*(p - q)| \leq \|x^*\| \|p - q\| \quad (p, q \in M).$$

Hence $\|Jx^*\|_{\text{Lip}_0} \leq \|x^*\|$.

Suppose $x^* \neq 0$. Reflexivity of X makes B_X weakly compact, so x^* attains its norm: after changing sign, there is $u \in S_X$ with $x^*(u) = \|x^*\|$. Lemma 1 gives $p, q \in M$ such that $p - q = u$. Therefore

$$\frac{|(Jx^*)(p) - (Jx^*)(q)|}{\|p - q\|} = \frac{|x^*(u)|}{\|u\|} = \|x^*\|.$$

Thus Jx^* strongly attains its Lipschitz norm at (p, q) and $\|Jx^*\|_{\text{Lip}_0} = \|x^*\|$. The map J is linear and isometric, so its range is closed because X^* is complete.

Now take $X = \ell_2$. Then $J(X^*)$ is isometric to ℓ_2 . An infinite-dimensional isomorphically polyhedral Banach space contains an isomorphic copy of c_0 [3]. This is impossible in the reflexive space ℓ_2 , because closed subspaces of reflexive spaces are reflexive whereas c_0 is not. Hence $J(\ell_2)$ is not isomorphically polyhedral, which disproves the assertion in Question 4.3. $\square$

4 Strength, limitations, and novelty check

The construction disproves exactly the printed question: that question assumes only that M is pointed and has the small ball property; it does not assume that M is complete. The set in (1) is a dense G_δ in X , but need not be complete for the inherited norm metric. Thus this packet does not decide the stronger variant obtained by adding metric completeness.

The separability half of the source conclusion is not contradicted: the displayed bad subspace is ℓ_2 , which is separable. The counterexample is decisive because the question asks for both separability and isomorphic polyhedrality.

On 11 August 2026, exact searches of the run registry and all cheap indexes found no prior packet or registered answer. Exact searches across the locally parsed arXiv corpus for the source title, the small ball property, strong norm attainment, and polyhedrality found the source and later papers discussing Questions 4.1–4.2, but no resolution of Question 4.3. Primary arXiv:2204.12529 answers Questions 4.1–4.2 by embedding c_0 in $\text{SNA}(M)$ for every infinite complete M [4]. Bounded web searches likewise returned the source and unrelated uses of “small ball.” This is a bounded novelty check, not an exhaustive priority claim.

A Source-page evidence

Figure 1 reproduces the source page containing Question 4.3.

References

- [1] V. Kadets and O. Roldán, *Closed linear spaces consisting of strongly norm attaining Lipschitz mappings*, arXiv:2202.06855v2 [math.FA], 2022; RACSAM 116 (2022), Paper 162.
- [2] E. Behrends and V. M. Kadets, *Metric spaces with the small ball property*, Studia Math. 148 (2001), 275–287.
- [3] V. P. Fonf, *Polyhedral Banach spaces*, Math. Notes 30 (1981), 809–813.
- [4] A. Avilés, G. Martínez-Cervantes, A. Rueda Zoca, and P. Tradacete, *Infinite dimensional spaces in the set of strongly norm-attaining Lipschitz maps*, arXiv:2204.12529 [math.FA], 2022; Rev. Mat. Iberoam. 40 (2024), 1409–1420.

All that remains is to note that if X is any metric space containing $[0, 1]$ isometrically, then the mapping F exists. Indeed, the identity operator Id on $[0, 1]$ is a Lipschitz function with constant 1, and by McShane's extension theorem, it can be extended to the whole X preserving its Lipschitz constant. $\square$

In particular, this holds for all normed spaces. This should be once more compared with the classical theory of norm-attaining functionals, where there exist Banach spaces X such that $\text{NA}(X, \mathbb{R})$ does not have 2-dimensional subspaces (see [25]).

4. OPEN QUESTIONS AND REMARKS

We have seen in Theorem 2.6 that if M is any infinite pointed metric space, then $\text{SNA}(M)$ contains linear subspaces of every possible finite dimension. However, we do not know if every infinite pointed metric space M satisfies that $\text{SNA}(M)$ contains infinite-dimensional linear subspaces, and Theorem 3.3 gives us some restrictions.

Question 4.1. *Is it true that for every infinite complete pointed metric space M the corresponding $\text{SNA}(M)$ contains infinite-dimensional closed (or at least non-closed) linear subspaces?*

Note that for Question 4.1, there is no chance to have isometric copies of ℓ_1 as a tool for its solution in general, since if the space M is separable, the candidates for linear subspaces of $\text{SNA}(M)$ need to have separable dual (see Theorem 3.3). Actually, in all the examples of infinite M that we were able to analyze in depth, $\text{SNA}(M)$ contains isomorphic copies of c_0 , but we do not know if this is true in general.

Question 4.2. *Is it true that for every infinite complete pointed metric space M the corresponding $\text{SNA}(M)$ contains an isomorphic copy of c_0 ?*

Let us comment that, as it was mentioned in Section 1.1, in [7, Theorem 3.2] it was shown that if M is an infinite metric space, then $\text{Lip}_0(M)$ contains a linear subspace isomorphic to ℓ_∞ , and an isometric version was achieved later in [8, Theorem 5], however, if one examines the proofs, there is no chance for them to be applicable to $\text{SNA}(M)$ in general, and actually, we refer once more to Theorem 3.3.

It has been shown in Theorem 3.7 that if M is a σ -precompact pointed metric space, then all subspaces of $\text{SNA}(M)$ are separable and isomorphically polyhedral. However, the authors do not know if the same holds for the weaker small ball property.

Question 4.3. *Let M be a pointed metric space with the small ball property. Is it true that all subspaces of $\text{SNA}(M)$ are separable and isomorphically polyhedral?*

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Figure 1: Printed page 10 of the source paper, including Question 4.3.